

It has long been understood that physical activity promotes health and longevity. Physical activity has been linked to dose-dependent reductions in a wide range of health conditions, such as cardiovascular disease and many cancers. However, with the advent of modern technology, human physical activity levels (PALs) have been decreasing in recent generations. PAL is defined as an individual's daily energy expenditure divided by that individual's basal metabolic rate (ie, daily energy expenditure without physical activity). Modern human adults have an average PAL of 1.6, while research suggests that ancestral humans in hunter-gatherer populations had an average PAL of 1.9, and ancestral and modern wild chimpanzees (humans' closest primate relatives) have an average PAL of 1.4.

Although it is understood that physical activity promotes health, there is not yet a cohesive evolutionary explanation for this. However, several hypotheses have been proposed to attempt to explain the evolutionary significance of physical activity on health and longevity. Two of these hypotheses are presented here:

Hypothesis 1

Energy spent performing physical activity reduces the fat storage that can compromise health in individuals.

Hypothesis 2

The stresses that performing physical activity put on the human body stimulate repair processes that preserve health. These processes are activated less in people who engage in less physical activity.

Which of the following conclusions is best supported by the information provided in the passage?

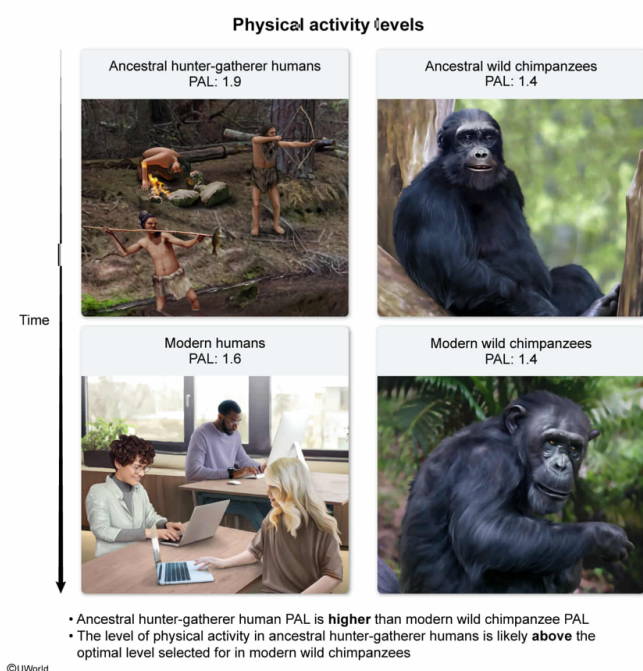
- ✔ ☒ A. The PAL of ancestral hunter-gatherer humans is likely above the optimal level selected for in modern wild chimpanzees. [76%]
- ☐ B. The PAL of modern wild chimpanzees is likely less than that of healthier ancestral wild chimpanzees. [15%]
- ☐ C. Ancestral hunter-gatherer humans likely had lower daily energy expenditures than did ancestral wild chimpanzees. [5%]
- ☐ D. Modern human adults likely expend more energy than did ancestral

hunter-gatherer humans. [2%]

Correct

75% answered correctly

Explanation:



Natural selection is a **mechanism of evolution** by which traits that are **adaptive** (ie, traits that increase **fitness**) in a particular environment are more likely to be passed on to the next generation than traits that are less favorable. Because of this, beneficial traits encoded by adaptive **alleles** become more common over time.

The passage describes PALs in humans and the fact that these have decreased in recent generations. PAL is defined as the daily energy expended by an individual divided by the individual's basal metabolic rate. The passage provides average PALs of modern human adults, ancestral hunter-gatherer humans, and ancestral and modern wild chimpanzees (humans' closest primate relatives).

Because the passage states that the average PAL of ancestral hunter-gatherer humans is **1.9** and the average PAL of ancestral and modern wild chimpanzees is **1.4**, it can be deduced that ancestral hunter-gatherer humans were **more active** than wild chimpanzees. Because beneficial traits become more common in populations, it is likely that the **level of physical activity** in **ancestral hunter-gatherer humans** is **above** the **optimal level selected for** in **modern wild chimpanzees**.

(Choice B) There is *no* information provided to suggest that ancestral wild chimpanzees were healthier than modern wild chimpanzees. In addition, since the passage states that the PAL of both ancestral and modern wild chimpanzees is the same (ie, 1.4), the level of physical activity in the two groups is likely *similar*.

(Choice C) The passage states that ancestral hunter-gatherer humans had an average PAL of 1.9, which is higher than the average ancestral wild chimpanzee PAL of 1.4. Therefore, ancestral hunter-gatherer humans likely had higher (*not* lower) daily energy expenditures than did ancestral wild chimpanzees.

(Choice D) Because the average PAL of modern human adults is less than the average PAL of ancestral hunter-gatherer humans, modern human adults likely expend less (*not* more) energy than did ancestral hunter-gatherer humans.

Educational objective:

During the process of natural selection, beneficial traits (ie, those that improve fitness) are more likely to be passed to subsequent generations than are less favorable traits.

Time spent:0 sec

Subject:
Biology

QID:404020

System:
Genetics and Evolution

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Which of the following could have facilitated the evolution of modern human adults from chimpanzee ancestors?

- I. Behavioral isolation
- II. Hybrid breakdown
- III. Convergent evolution

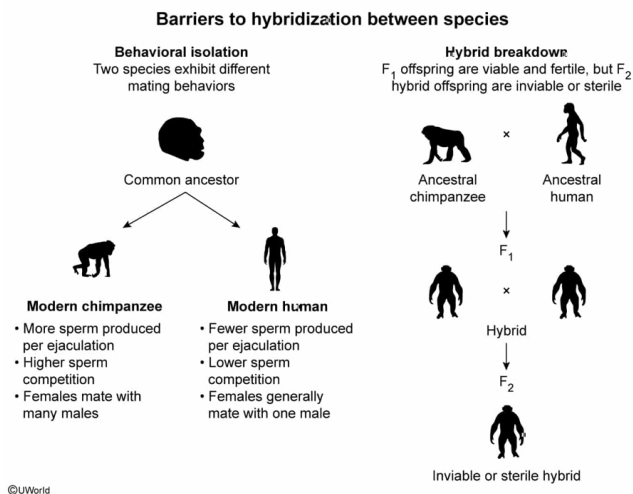
- ☐ A. I only [25%]
- ☒ B. I and II only [40%]
- ☐ C. I and III only [23%]

○ D. II and III only [10%]

Correct

40% answered correctly

Explanation:



New **species** can form during the process of **speciation**, and all current species are thought to come from species that already existed. The **biological concept** of a species is a group of organisms able to **sexually reproduce** to form fertile offspring.

Barriers to hybridization (ie, reproductive isolation) keep species separated. These barriers can be broadly separated into two groups:

- **Prezygotic barriers** (ie, barriers occurring **before zygote formation**)
 - Temporal isolation: Organisms of different species may reproduce at different times.
 - **Ecological isolation**: Organisms of different species may not have geographic access to one another.
 - Behavioral isolation: Organisms of different species may display different mating or courtship behaviors (**Number I**).
 - Mechanical isolation: Organisms of different species may not have compatible reproductive organs.
 - Gametic isolation: The sperm of one species may not recognize the egg of another species.

- **Postzygotic barriers** (ie, barriers occurring **after** zygote formation)
 - Hybrid sterility: The organism produced by the mating of different species may not be able to produce fertile offspring itself.
 - **Hybrid inviability**: The organism produced by the mating of different species may not develop normally, leading to death prior to reproduction of the hybrid.
 - Hybrid breakdown: The hybrid second generation produced by the successful mating of two hybrid organisms may not be biologically healthy (**Number II**).

(Number III) **Convergent evolution** occurs when **distantly related** organisms become *more* alike by evolving similar traits in an independent manner due to similar needs. For example, fish and dolphins are distantly related species that have both evolved the trait of fins due to their similar aquatic environment. However, speciation involves a change that causes organisms originally within a single species to become reproductively *less* alike (ie, reproductively isolated). The passage states that chimpanzees are humans' closest primate relatives. Therefore, convergent evolution would result in *greater* overall similarity between the two species, without any necessary effect on their reproductive compatibility.

Educational objective:

A biological species is a group of organisms able to sexually reproduce and form fertile offspring, and speciation is the formation of new species.

Time spent:0 sec

Subject:
Biology

QID:404021

System:
Genetics and Evolution

It has long been understood that physical activity promotes health and longevity. Physical activity has been linked to dose-dependent reductions in a wide range of health conditions, such as cardiovascular disease and many cancers. However, with the advent of modern technology, human physical activity levels (PALs) have been decreasing in recent generations. PAL is defined as an individual's daily energy expenditure divided by that individual's basal metabolic rate (ie, daily energy expenditure without physical activity). Modern human adults have an average PAL of 1.6, while research suggests that ancestral humans in hunter-gatherer populations had an average PAL of 1.9, and ancestral and modern wild chimpanzees (humans' closest primate relatives) have an average PAL of 1.4.

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Hypothesis 2

The stresses that performing physical activity put on the human body stimulate repair processes that preserve health. These processes are activated less in people who engage in less physical activity.

When comparing two modern human populations to test the passage hypotheses, similarity in what respect between the two populations is most important?

- ☐ A. Physical activity level [43%]
- ☐ B. Average lifespan [15%]

☐ C. Geographic location [18%]

✓ ☒ D. Population size [22%]

Correct

23% answered correctly

Explanation:

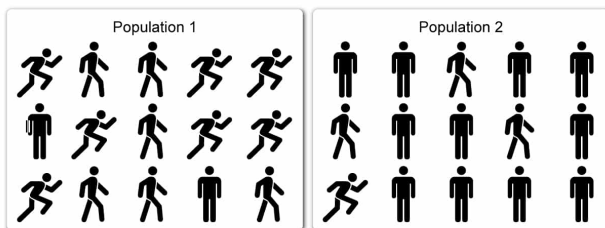
Controlling variables in an experiment

- Hypothesis 1: Energy spent performing physical activity reduces detrimental fat storage.
- Hypothesis 2: Physical activity stress stimulates repair processes that preserve health.

1. Select similarly-sized populations (**control variable**)



Increasing PAL (**independent variable**)



2. Determine PALs in population members over time
3. Measure average lifespan in each population (**dependent variable**) to test hypotheses
4. Draw conclusions

PAL = physical activity level.

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In an experiment, **control variables** are used to demonstrate that any changes observed in the **dependent variable** (ie, what is being measured) are due only to changes in the **independent variable** (ie, what is being manipulated). This helps ensure that experimental results are not impacted by **confounding variables**.

The passage presents two hypotheses about the evolutionary significance of physical activity on health. If two modern human **populations** are compared to test these hypotheses, certain variables should be controlled so that what is measured is due only to PAL, rather than a confounding variable. One example of a control variable in this case would be **population size**. Comparing populations of two different sizes would likely introduce other variables into the experiment that could alter the results. Therefore, comparing **two populations of a similar size** would be important to producing accurate experimental data.

(Choice A) The passage hypotheses center around the effect of PAL. Therefore, PAL is the independent (*not* control) variable in this situation and should therefore *differ* between the populations.

(Choice B) The average lifespan of each population is what would be measured; therefore, this is the *dependent* variable in the experiment.

(Choice C) If the humans being measured are in the same geographic location, they would be part of the *same* population, not different populations.

Educational objective:

Certain variables are controlled in experiments to demonstrate that any changes in the dependent variable are due only to the manipulation of the independent variable by the experimenter, rather than by confounding variables.

Time spent:0 sec

Subject:
Biology

QID:404022

System:
Genetics and Evolution

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Hypothesis 2

The stresses that performing physical activity put on the human body stimulate repair processes that preserve health. These processes are activated less in people who engage in less physical activity.

Research has shown that modern human lifespans are approximately 10 years longer than those of ancestral hunter-gatherer humans, yet modern human adults engage in much lower levels of physical activity. Given this, which statement best explains the longer lifespans of modern human adults?

- ☒ A. Diseases are managed differently in modern human adults than they were in ancestral hunter-gatherer humans. [82%]
- ☐ B. Modern human adults engage in different types of physical activity than did ancestral hunter-gatherer humans. [6%]
- ☐ C. Modern human adults contain less body fat than did ancestral hunter-

gatherer humans. [4%]

- ☐ D. Physical activity puts less stress on modern human adults than it did on ancestral hunter-gatherer humans. [6%]

Correct

83% answered correctly

Explanation:

Longevity in modern humans versus ancestral hunter-gatherer humans



Natural selection acts on alleles that increase fitness in a certain environment. New alleles occur via mutations. All things being equal, advantageous mutations (ie, those increasing fitness) will increase in a population, while deleterious mutations (ie, those decreasing fitness) will decrease in a population over time.

The overall effects of beneficial and deleterious mutations occur against the backdrop of a population's environment. Environmental factors (eg, water and food availability/cleanliness, diet, vaccinations, mother/infant care, wars) can supersede or mask the effects of genetic variation, making consideration of both genes *and* environment crucial in assessment of lifespan changes.

In this question, research has shown that modern human lifespans are approximately 10 years longer than those of ancestral hunter-gatherer humans, but modern human adults engage in much lower levels of physical activity. The passage states that increased physical activity is associated with increased lifespan.

One possible explanation for this unexpected phenomenon is the **difference in disease management in modern human adults**. Although PALs (and therefore, overall health) are lower in modern times, access to modern healthcare to manage diseases can prolong lifespans in unhealthy individuals. In ancestral hunter-gatherer humans, longer lifespans would have likely come about due to overall good health.

(Choice B) While differences in types of physical activity (eg, running on a treadmill, hunting for food) likely exist between modern and ancestral hunter-gatherer humans, there is no information given about type of physical activity

and its contribution to overall health. Therefore, this is *not* the best choice to explain the lifespans in modern human adults versus ancestral hunter-gatherer humans.

(Choice C) Energy spent performing physical activity may reduce fat storage (Hypothesis 1). Therefore, because modern human adults have lower levels of physical activity, modern human adults likely have more (*not* less) body fat than ancestral hunter-gatherer humans.

(Choice D) Hypothesis 2 posits that physical activity-related stresses lead to the preservation of health. Therefore, if physical activity puts less stress on modern human adults than it did on ancestral hunter-gatherer humans, this would likely lead to shorter (*not* longer) lifespans in modern human adults compared to ancestral hunter-gatherer humans.

Educational objective:

As a result of natural selection, advantageous traits (ie, those improving fitness) are likely to increase over time, while deleterious traits (ie, those decreasing fitness) are likely to decrease over time. Environmental factors may affect which traits are effective, and to what extent.

Time spent:0 sec

Subject:

Biology

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System:

Genetics and Evolution